

# Montrac Robot Operations

## Manual

Description of robotic operations and their parameters.

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W1/W2 Montrac

V4

2026-02-03

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## CONTENTS

|                                              |    |
|----------------------------------------------|----|
| Operation - ID 10 - MOVE TO HOME .....       | 2  |
| Operation - ID 11 - MOVE LIN CARTESIAN ..... | 3  |
| Operation - ID 12 - MOVE PTP CARTESIAN ..... | 4  |
| Operation - ID 13 - MOVE PTP AXIS .....      | 5  |
| Operation - ID 20 - PICK .....               | 6  |
| Operation - ID 21 - PICK VERTICAL .....      | 7  |
| Operation - ID 30 - PLACE .....              | 8  |
| Operation - ID 31 - PLACE VERTICAL .....     | 9  |
| Operation - ID 40 - APPROACH .....           | 10 |
| Operation - ID 41 - APPROACH VERTICAL .....  | 11 |
| Operation - ID 50 - BRAKE TEST .....         | 12 |
| Operation - ID 51 - RUN TOOL COMMAND .....   | 13 |
| Operation - ID 200 - PICK PTP .....          | 14 |
| Operation - ID 201 - PICK PTP RSI .....      | 15 |

## OPERATION - ID 10 - MOVE TO HOME

Moves the robot into its home position in PTP motion.

| Table of operation parameters |       |          |                                               |
|-------------------------------|-------|----------|-----------------------------------------------|
| Index                         | Name  | Values   | Description                                   |
| 0                             | Speed | 1 to 100 | Percentual speed override of the robot motion |

### Robot actions:

1. The robot **moves to its home position** in **PTP motion**.

## OPERATION - ID 11 - MOVE LIN CARTESIAN

Moves the robot into a cartesian position using a linear motion.

| Table of operation parameters |         |                         |                                                       |
|-------------------------------|---------|-------------------------|-------------------------------------------------------|
| Index                         | Name    | Values                  | Description                                           |
| 0                             | X       | -5,000,000 to 5,000,000 | X position in micrometers                             |
| 1                             | Y       | -5,000,000 to 5,000,000 | Y position in micrometers                             |
| 2                             | Z       | -5,000,000 to 5,000,000 | Z position in micrometers                             |
| 3                             | A       | -360,000 to 360,000     | Rotation along Z axis in millidegrees                 |
| 4                             | B       | -360,000 to 360,000     | Rotation along Y axis in millidegrees                 |
| 5                             | C       | -360,000 to 360,000     | Rotation along X axis in millidegrees                 |
| 6                             | Speed   | 1 to 100                | Percentual speed override of the robot motion         |
| 7                             | Base ID | 0 to 127                | ID of the base frame in which the motion will be done |

### Robot actions:

1. The robot **moves to the defined position** in **LIN motion**.

## OPERATION - ID 12 - MOVE PTP CARTESIAN

Moves the robot into a cartesian position using a PTP motion.

| Table of operation parameters |         |                         |                                                       |
|-------------------------------|---------|-------------------------|-------------------------------------------------------|
| Index                         | Name    | Values                  | Description                                           |
| 0                             | X       | -5,000,000 to 5,000,000 | X position in micrometers                             |
| 1                             | Y       | -5,000,000 to 5,000,000 | Y position in micrometers                             |
| 2                             | Z       | -5,000,000 to 5,000,000 | Z position in micrometers                             |
| 3                             | A       | -360,000 to 360,000     | Rotation along Z axis in millidegrees                 |
| 4                             | B       | -360,000 to 360,000     | Rotation along Y axis in millidegrees                 |
| 5                             | C       | -360,000 to 360,000     | Rotation along X axis in millidegrees                 |
| 6                             | Status  | 0 to 7                  | KUKA Position Status parameter                        |
| 7                             | Turn    | 0 to 63                 | KUKA Position Turn parameter                          |
| 8                             | Speed   | 1 to 100                | Percentual speed override of the robot motion         |
| 9                             | Base ID | 0 to 127                | ID of the base frame in which the motion will be done |

### Robot actions:

1. The robot **moves to the defined position** in **PTP motion**.

## OPERATION - ID 13 - MOVE PTP AXIS

Moves the robot into an axis specific position using a PTP motion.

| Table of operation parameters |       |                     |                                               |
|-------------------------------|-------|---------------------|-----------------------------------------------|
| Index                         | Name  | Values              | Description                                   |
| 0                             | A1    | -360,000 to 360,000 | Axis position millidegrees                    |
| 1                             | A2    | -360,000 to 360,000 | Axis position millidegrees                    |
| 2                             | A3    | -360,000 to 360,000 | Axis position millidegrees                    |
| 3                             | A4    | -360,000 to 360,000 | Axis position millidegrees                    |
| 4                             | A5    | -360,000 to 360,000 | Axis position millidegrees                    |
| 5                             | A6    | -360,000 to 360,000 | Axis position millidegrees                    |
| 6                             | Speed | 1 to 100            | Percentual speed override of the robot motion |

### Robot actions:

1. The robot **moves to the defined position** in **PTP motion**.

## OPERATION - ID 20 - PICK

Performs a picking motion with the robot and a gripper. All motions are linear.

| Table of operation parameters |                       |                         |                                                                                              |
|-------------------------------|-----------------------|-------------------------|----------------------------------------------------------------------------------------------|
| Index                         | Name                  | Values                  | Description                                                                                  |
| 0                             | X                     | -5,000,000 to 5,000,000 | X position in micrometers                                                                    |
| 1                             | Y                     | -5,000,000 to 5,000,000 | Y position in micrometers                                                                    |
| 2                             | Z                     | -5,000,000 to 5,000,000 | Z position in micrometers                                                                    |
| 3                             | A                     | -360,000 to 360,000     | Rotation along Z axis in millidegrees                                                        |
| 4                             | B                     | -360,000 to 360,000     | Rotation along Y axis in millidegrees                                                        |
| 5                             | C                     | -360,000 to 360,000     | Rotation along X axis in millidegrees                                                        |
| 6                             | Speed                 | 1 to 100                | Percentual speed override of the robot motion                                                |
| 7                             | Base ID               | 0 to 127                | ID of the base frame in which the motion will be done                                        |
| 8                             | Return                | 0 or 1                  | If the robot should return to its starting position after the motion                         |
| 9                             | Offset                | 100,000 to 500,000      | Offset of the approach position from the provided position in the Z direction in micrometers |
| 10                            | Gripper Open Position | -100,000 to 100,000     | Position of the gripper jaws in micrometers, if allowed by the gripper type                  |

### Robot actions:

1. The gripper is moved to the specified position.
2. The robot moves to an approach position, offset from the provided gripping position by the provided offset in the Z direction.
3. The robot moves to the specified gripping position.
4. The gripper is closed.
5. The robot moves back to the approach position.
6. OPTIONAL: The robot returns to its starting position.

## OPERATION - ID 21 - PICK VERTICAL

Performs a picking motion with the robot and a gripper. All motions are linear. The gripper is pointing down vertically during the motion, with respect to the base frame. The only rotation which needs to be provided, is along the Z axis.

| Table of operation parameters |                       |                         |                                                                                              |
|-------------------------------|-----------------------|-------------------------|----------------------------------------------------------------------------------------------|
| Index                         | Name                  | Values                  | Description                                                                                  |
| 0                             | X                     | -5,000,000 to 5,000,000 | X position in micrometers                                                                    |
| 1                             | Y                     | -5,000,000 to 5,000,000 | Y position in micrometers                                                                    |
| 2                             | Z                     | -5,000,000 to 5,000,000 | Z position in micrometers                                                                    |
| 3                             | A                     | -360,000 to 360,000     | Rotation along Z axis in millidegrees                                                        |
| 4                             | Speed                 | 1 to 100                | Percentual speed override of the robot motion                                                |
| 5                             | Base ID               | 0 to 127                | ID of the base frame in which the motion will be done                                        |
| 6                             | Return                | 0 or 1                  | If the robot should return to its starting position after the motion                         |
| 7                             | Offset                | 100,000 to 500,000      | Offset of the approach position from the provided position in the Z direction in micrometers |
| 8                             | Gripper Open Position | -100,000 to 100,000     | Position of the gripper jaws in micrometers, if allowed by the gripper type                  |

### Robot actions:

1. The gripper is moved to the specified position.
2. The robot moves to an approach position, offset from the provided gripping position by the provided offset in the Z direction.
3. The robot moves to the specified gripping position.
4. The gripper is closed.
5. The robot moves back to the approach position.
6. OPTIONAL: The robot returns to its starting position.

## OPERATION - ID 30 - PLACE

Performs a placing motion with the robot and a gripper. All motions are linear.

| Table of operation parameters |                       |                         |                                                                                              |
|-------------------------------|-----------------------|-------------------------|----------------------------------------------------------------------------------------------|
| Index                         | Name                  | Values                  | Description                                                                                  |
| 0                             | X                     | -5,000,000 to 5,000,000 | X position in micrometers                                                                    |
| 1                             | Y                     | -5,000,000 to 5,000,000 | Y position in micrometers                                                                    |
| 2                             | Z                     | -5,000,000 to 5,000,000 | Z position in micrometers                                                                    |
| 3                             | A                     | -360,000 to 360,000     | Rotation along Z axis in millidegrees                                                        |
| 4                             | B                     | -360,000 to 360,000     | Rotation along Y axis in millidegrees                                                        |
| 5                             | C                     | -360,000 to 360,000     | Rotation along X axis in millidegrees                                                        |
| 6                             | Speed                 | 1 to 100                | Percentual speed override of the robot motion                                                |
| 7                             | Base ID               | 0 to 127                | ID of the base frame in which the motion will be done                                        |
| 8                             | Return                | 0 or 1                  | If the robot should return to its starting position after the motion                         |
| 9                             | Offset                | 100,000 to 500,000      | Offset of the approach position from the provided position in the Z direction in micrometers |
| 10                            | Gripper Open Position | -100,000 to 100,000     | Position of the gripper jaws in micrometers, if allowed by the gripper type                  |

### Robot actions:

1. The robot moves to a point defined by the specified coordinates, offset by the provided offset in the Z direction.
2. The robot moves directly to the specified cartesian placing point.
3. The gripper is opened to the specified position.
4. The robot moves back to the offset point.
5. OPTIONAL: The robot returns to its starting position.



## OPERATION - ID 31 - PLACE VERTICAL

Performs a placing motion with the robot and a gripper. All motions are linear. The gripper is pointing down vertically during the motion, with respect to the base frame. The Only rotation which needs to be provided, is along the Z axis.

| Table of operation parameters |                       |                         |                                                                                              |
|-------------------------------|-----------------------|-------------------------|----------------------------------------------------------------------------------------------|
| Index                         | Name                  | Values                  | Description                                                                                  |
| 0                             | X                     | -5,000,000 to 5,000,000 | X position in micrometers                                                                    |
| 1                             | Y                     | -5,000,000 to 5,000,000 | Y position in micrometers                                                                    |
| 2                             | Z                     | -5,000,000 to 5,000,000 | Z position in micrometers                                                                    |
| 3                             | A                     | -360,000 to 360,000     | Rotation along Z axis in millidegrees                                                        |
| 4                             | Speed                 | 1 to 100                | Percentual speed override of the robot motion                                                |
| 5                             | Base ID               | 0 to 127                | ID of the base frame in which the motion will be done                                        |
| 6                             | Return                | 0 or 1                  | If the robot should return to its starting position after the motion                         |
| 7                             | Offset                | 100,000 to 500,000      | Offset of the approach position from the provided position in the Z direction in micrometers |
| 8                             | Gripper Open Position | -100,000 to 100,000     | Position of the gripper jaws in micrometers, if allowed by the gripper type                  |

### Robot actions:

1. The robot moves to a point defined by the specified coordinates, offset by the provided offset in the Z direction.
2. The robot moves directly to the specified cartesian placing point.
3. The gripper is opened to to the specified position.
4. The robot moves back to the offset point.
5. OPTIONAL: The robot returns to its starting position.

## OPERATION - ID 40 - APPROACH

Approaches a specified position with the robot. All motions are linear.

| Table of operation parameters |         |                         |                                                                                              |
|-------------------------------|---------|-------------------------|----------------------------------------------------------------------------------------------|
| Index                         | Name    | Values                  | Description                                                                                  |
| 0                             | X       | -5,000,000 to 5,000,000 | X position in micrometers                                                                    |
| 1                             | Y       | -5,000,000 to 5,000,000 | Y position in micrometers                                                                    |
| 2                             | Z       | -5,000,000 to 5,000,000 | Z position in micrometers                                                                    |
| 3                             | A       | -360,000 to 360,000     | Rotation along Z axis in millidegrees                                                        |
| 4                             | B       | -360,000 to 360,000     | Rotation along Y axis in millidegrees                                                        |
| 5                             | C       | -360,000 to 360,000     | Rotation along X axis in millidegrees                                                        |
| 6                             | Speed   | 1 to 100                | Percentual speed override of the robot motion                                                |
| 7                             | Base ID | 0 to 127                | ID of the base frame in which the motion will be done                                        |
| 8                             | Return  | 0 or 1                  | If the robot should return to its starting position after the motion                         |
| 9                             | Offset  | 100,000 to 500,000      | Offset of the approach position from the provided position in the Z direction in micrometers |

### Robot actions:

1. The robot moves to an approach position, offset from the provided position by the provided offset in the Z direction.
2. The robot moves to the specified position.
3. The robot moves back to the approach position.
4. OPTIONAL: The robot returns to its starting position.

## OPERATION - ID 41 - APPROACH VERTICAL

Approaches a specified position with the robot. All motions are linear. The gripper is pointing down vertically during the motion, with respect to the base frame. The only rotation which needs to be provided, is along the Z axis.

| Table of operation parameters |         |                         |                                                                                              |
|-------------------------------|---------|-------------------------|----------------------------------------------------------------------------------------------|
| Index                         | Name    | Values                  | Description                                                                                  |
| 0                             | X       | -5,000,000 to 5,000,000 | X position in micrometers                                                                    |
| 1                             | Y       | -5,000,000 to 5,000,000 | Y position in micrometers                                                                    |
| 2                             | Z       | -5,000,000 to 5,000,000 | Z position in micrometers                                                                    |
| 3                             | A       | -360,000 to 360,000     | Rotation along Z axis in millidegrees                                                        |
| 4                             | Speed   | 1 to 100                | Percentual speed override of the robot motion                                                |
| 5                             | Base ID | 0 to 127                | ID of the base frame in which the motion will be done                                        |
| 6                             | Return  | 0 or 1                  | If the robot should return to its starting position after the motion                         |
| 7                             | Offset  | 100,000 to 500,000      | Offset of the approach position from the provided position in the Z direction in micrometers |

### Robot actions:

1. The robot moves to an approach position, offset from the provided position by the provided offset in the Z direction.
2. The robot moves to the specified position.
3. The robot moves back to the approach position.
4. OPTIONAL: The robot returns to its starting position.

## **OPERATION - ID 50 - BRAKE TEST**

Performs brake test on the robot, if it is requested by the robot controller.

### **Robot actions:**

1. The robot performs a brake test.

## OPERATION - ID 51 - RUN TOOL COMMAND

Commands the gripper currently attached to the robot to perform to perform a gripper operation, if possible.

| Table of operation parameters |                 |           |                                     |
|-------------------------------|-----------------|-----------|-------------------------------------|
| Index                         | Name            | Values    | Description                         |
| 0                             | Gripper Command | 1 to 1000 | ID of the requested gripper command |
| 1                             | Gripper Command | -         | Parameter 1                         |

### Robot actions:

1. The attached gripper performs the specified operation.

List of available gripper commands:

| Gripper Command Number | Name                |
|------------------------|---------------------|
| 10                     | Acknowledge Error   |
| 11                     | Reference           |
| 20                     | Release             |
| 21                     | Release to position |
| 30                     | Grip                |

## OPERATION - ID 200 - PICK PTP

| Table of operation parameters |        |                     |                                                                                                      |
|-------------------------------|--------|---------------------|------------------------------------------------------------------------------------------------------|
| Index                         | Name   | Values              | Description                                                                                          |
| 0                             | A1     | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                               |
| 1                             | A2     | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                               |
| 2                             | A3     | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                               |
| 3                             | A4     | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                               |
| 4                             | A5     | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                               |
| 5                             | A6     | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                               |
| 6                             | Speed  | 1 to 100            | Percentual speed override of the robot<br>motion                                                     |
| 7                             | Offset | 100,000 to 500,000  | Offset of the approach position from the<br>provided position in the Z direction in mi-<br>crometers |

### Robot actions:

1. The **gripper** is **opened**.
2. The **Base ID** is **set to 1**.
3. The **robot** moves to the provided axis **pre-pick position** in a **PTP motion**.
4. The **robot** moves a **gripping position**, which is offset from the provided axis position in the tool Z direction in a **LIN motion**.
5. The **gripper** is **closed**.
6. The **robot** moves back to the **pre-pick position** in a **LIN motion**.
7. The **robot** moves to a **hardcoded position** in a **PTP motion**.

## OPERATION - ID 201 - PICK PTP RSI

| Table of operation parameters |                                 |                     |                                                                                                    |
|-------------------------------|---------------------------------|---------------------|----------------------------------------------------------------------------------------------------|
| Index                         | Name                            | Values              | Description                                                                                        |
| 0                             | A1                              | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                             |
| 1                             | A2                              | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                             |
| 2                             | A3                              | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                             |
| 3                             | A4                              | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                             |
| 4                             | A5                              | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                             |
| 5                             | A6                              | -360,000 to 360,000 | Pre-Pick Axis position<br>millidegrees                                                             |
| 6                             | Speed                           | 1 to 100            | Percentual speed override of the robot<br>motion                                                   |
| 7                             | Offset                          | 100,000 to 500,000  | Offset of the approach position from<br>the provided position in the Z direction<br>in micrometers |
| 8                             | Encoder<br>Captured<br>Position | -                   | Decimillimeters                                                                                    |

### Robot actions:

1. The **gripper** is **opened**.
2. The **Base ID** is **set to 1**.
3. The **robot** moves to a **hardcoded position** in a **PTP motion**.
4. The **robot** activates **RSI** correction.
5. The **robot** moves to the provided axis **pre-pick position** in a **PTP motion**.
6. The **robot** waits for a go signal from **RSI**.
7. The **robot** moves a **gripping position**, which is offset from the provided axis position in the tool Z direction in a **LIN motion**.
8. The **gripper** is **closed**.
9. The **robot** moves back to the **pre-pick position** in a **LIN motion**.
10. The **robot** deactivates **RSI** correction.
11. The **robot** moves to a **hardcoded position** in a **PTP motion**.